1. Systems Programming 2

School of Computer Science

Computer Science Department

Course no.: 2-7023010-1

Degree: B.A.

2. Schedule

2nd year, 2nd semester.

3. Lecturer

Dr. Elina Opalinski
Office hours: Upon appointment.

Lecturer in charge: Prof. Erel Segal-Halevi, 58.3.26, 09-7431290, davidesh@ariel.ac.il.
Office hours: Tuesday 11:00-13:00, upon appointment.

4. Teaching assistant/tutor

TA in charge (assignment manager): Avi Rahimov.

5. Course goal

The goal of the course is to train the students in learning a particularly complex and difficult programming language – C++. A language combining different programming paradigms --- procedural programming, object-oriented programming, functional programming and generic template programming. A language that requires manual handling of memory resources.

A secondary goal is to train the students in skills that are vital for any programmer: programming in a Linux environment, applying bash scripts, using git, object oriented design, clean code and code review.

6. Prerequisites

Passing grades in Systems Programming 1 (C programming).

7. Method of Instruction

- A weekly lecture (3 hours).
- A weekly tutorial class (1 hour), dedicated to rehearsing and reinforcing the material taught in the lecture. Given by a TA.
- A weekly *laboratory class* (1 hour), dedicated to code review and evaluation of the students' homework assignments. Given by a TA.

8. Course plan

The plan is tentative and subject to change.

Lesson unit	Subject	Lesson #
1	Introduction: C++ vs. C and Java. Namespaces, overloading, exceptions. Valgrind, tidy, doctest.	1
2	Classes and objects, construction and destruction	2
3	Composition, initialization list. References vs. pointers. Const, Const methods. Mutable.	3
4	Friend; operator overloading.	4
5	Deep copy; copy constructor and assignment operator. Conversion constructor and conversion operator. Rule of the three.	5
6	Inheritance: simple and virtual.	6
7	Templates and iterators.	7
8	Programming in templates. Functors and lambda expressions.	8
9	Standard library: containers and iterators.	9
10	Standard library: algorithms.	10
11	Smart pointers.	11
12	Rehearsal; Leetcode questions.	12

9. Course requirements:

There are 4 assignments, all obligatory. Each worth 5% of the final grade.

Additionally, each student must come to a laboratory twice per semester, for code review. During the laboratory, the TA will randomly choose one of two previously-submitted assignments (1 and 2, or 3 and 4) and examine the student's knowledge regarding that assignment. Each such exam is worth 10% of the final grade.

Finally, the final exam is worth 60% of the final grade.

10. Learning Outcomes:

By the end of the course, you will be able to:
1. Evaluate the differences between C++, C and Java; provide advantages and disadvantages of each language in each scenario.
2. Write software projects in C++.
3. Correctly use namespaces, operators, templates and iterators when they are needed.
4. Correctly use the standard C++ library when it is needed.
5. Compute the virtual table of classes in a virtual inheritance scenario.

11. Main textbook and additional textbooks:

1. Lecture notes from previous years (in Hebrew) are available at GitHub: https://github.com/erelsgl-at-ariel/cpp-5785

2. A Tour of C++ (2nd Edition), by Bjarne Stroustrup,

3. C++17 STL Cookbook: Discover the latest enhancements to functional programming and lambda expressions. by Jacek Galowicz. Paperback, June 28, 2017.

12. Required material for the examination

All material taught during class, including student presentations, and all assignments.